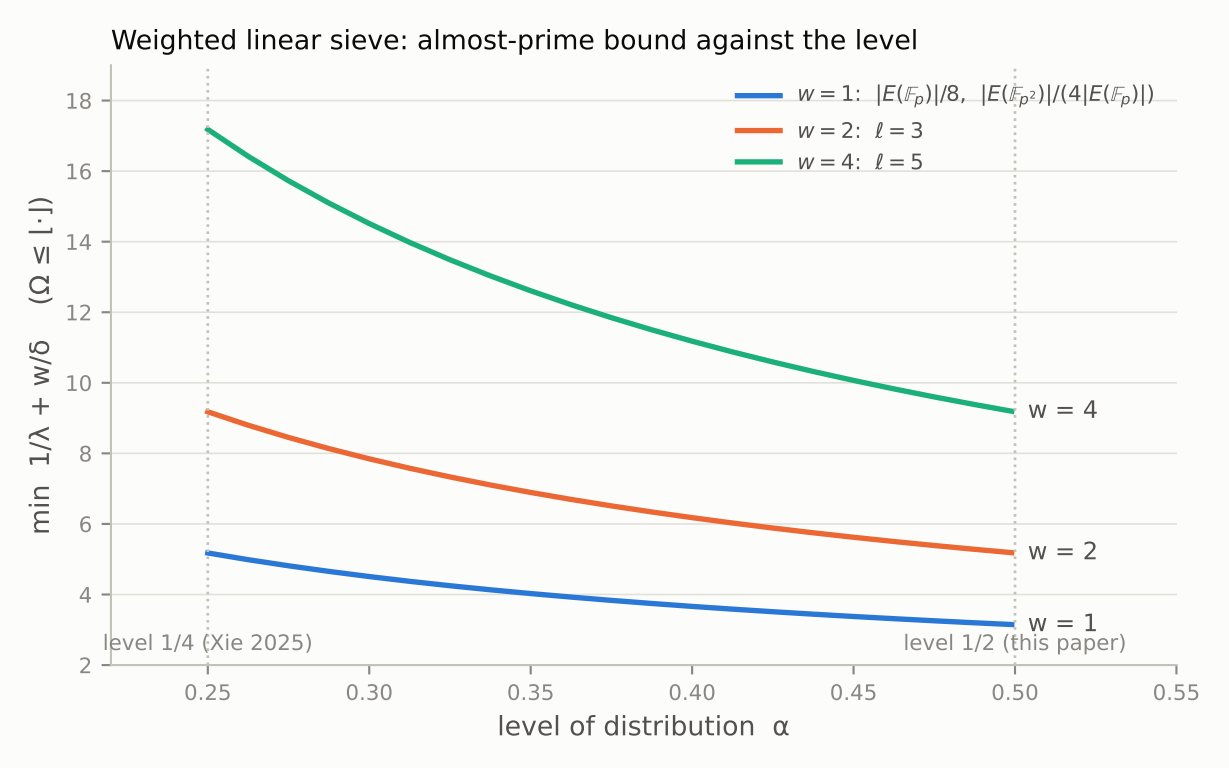
Weighted linear sieve: almost-prime bound against the level
$w = 1$: $|E(\mathbb{F}_p)|/8$, $|E(\mathbb{F}_{p^2})|/(4|E(\mathbb{F}_p)|)$
$w = 2$: $\ell = 3$
$w = 4$: $\ell = 5$
w = 4
w = 2
w = 1
level 1/4 (Xie 2025)
level 1/2 (this paper)
min 1/λ + w/δ (Ω ≤ ⌊·⌋)
level of distribution α
2
4
6
8
10
12
14
16
18
0.25
0.30
0.35
0.40
0.45
0.50
0.55